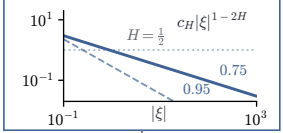
fixed basis

covariance norm

$$\|f\|_H = \|U_H f\|_{L^2(\mathbb{R})}$$

$U_H f = \sqrt{c_H}\,|\xi|^{1/2-H}\,\mathcal{F}f$

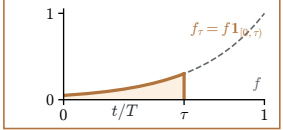


admits f

input: samples or integrals

any f with $\|f\|_H < \infty$

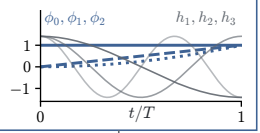
same rule for every f



raw nested sequence

$$\phi_0, \phi_1, \phi_2, \phi_3, \dots$$

$T^{-1/2},\ t/T,\ (t/T)^2,\ h_{k\geq 1}$

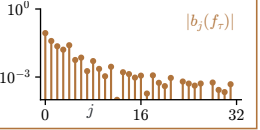


ϕ_j

moments

$$b_D(f) = (\langle f, \phi_j\rangle_H)_{j<D}$$

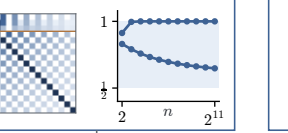
endpoint-corrected DCT



Gram, factored once

$$A_D = C_D C_D^\top$$

$\frac{1}{2}W \preceq S_+ \preceq W$ (cosines)

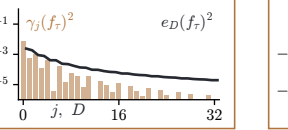


C_D

fixed coefficient rule

$$\gamma = C_D^{-1} b_D(f)$$

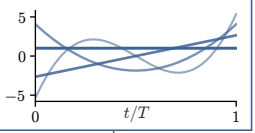
$e_D(f)^2 = \|f\|_H^2 - |\gamma|^2$



orthonormal, nested in D

$$v = C_D^{-1}\phi$$

$\langle v_i, v_j\rangle_H = \delta_{ij}$

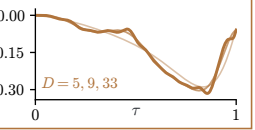


$Z_j = I_H(v_j)$

Wiener integral

$$I_H(f) \approx \textstyle\sum_{j<D} \gamma_j Z_j$$

one $Z \sim N(0, I_D)$ for all f



each input